Boundary-uniform relative smoothness and unique-minimizer convergence of B-adaPG

Abstract

We prove convergence of the original undamped B-adaPG method when its convex objective has a unique minimizer and relative smoothness is uniform on bounded interior sets approaching the kernel boundary. No strict convexity on the feasible closure is required of the kernel, the finite proximal term, their sum, or the objective. A retained smooth defect in the exact energy inequality yields a scalar separation principle and a cumulative-stepsize excursion argument. All cluster gradients then agree, allowing a unique-minimizer endpoint trap for the reverse Bregman distance. Large-step endpoints and descent at small steps give objective convergence and finally point convergence. Initialization is arbitrary and the full ambient controller is unchanged. An affine proximal term and nonlinear forward term illustrate the result on a flat boundary kernel. The stronger smoothness hypothesis remains essential to this proof; the corresponding question under only compact-interior relative smoothness is not settled here. No novelty claim is made.

Keywords: adaptive Bregman proximal gradient; boundary-uniform relative smoothness; unique minimizer; flat boundary kernels; Bregman distance; point convergence

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1 Setting and exact controller

For a differentiable convex function v, write $D_v(a,b) = v(a) - v(b) - \langle \nabla v(b), a-b\rangle$, with the second argument in its differentiability domain. The conjugate is $h^*(p) = \sup_x\{\langle p, x\rangle - h(x)\}$. We use the Euclidean norm. A function is 1-coercive if $h(x)/\|x\| \to +\infty$ as $\|x\| \to \infty$. Legendre means essentially smooth and essentially strictly convex.

We study the original undamped B-adaPG controller of Ou, Latafat and Themelis [1], pinned to arXiv:2508.01353v2. Our added assumptions are stated explicitly below. The proof uses scalar smooth-defect separation and objective convergence, rather than strict boundary geometry of the kernel.

1.1 Hypotheses

Let $h : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper, lower semicontinuous, convex, 1-coercive and Legendre. Let $C = \operatorname{int}\operatorname{dom} h \neq \varnothing$ and $K = \operatorname{dom} h = \overline{C}$. Assume $h \in C^2(C)$ with positive definite Hessian, and the following boundary properties (Assumption 2.4 of the cited paper):

1. $D_h(z, y_i) \to 0$ whenever $z \in K$, $y_i \in C$, and $y_i \to z$.

2. For every $z \in K \setminus C$, the map $y \mapsto D_h(z, y)$ is level bounded.

No strict convexity of h on K or on the feasible closure is assumed.

Let $U \supset K$ be open and convex and $f : U \to \mathbb{R}$ be convex and continuously differentiable. Require the following stronger-than-source relative smoothness:

For every $R > 0$ there exists $L_R > 0$ such that $L_R h - f$ is convex on $C \cap B(0, R)$.

Thus the modulus is uniform up to boundary approaches inside a fixed bounded ball. Let $g_0 : \mathbb{R}^n \to \mathbb{R}$ be finite convex, and D be closed convex with $D \cap C \neq \varnothing$. Write $S = K \cap D$ and $F = f + g_0$ on S. Suppose F attains its minimum at exactly one point z. No strict convexity of h, g_0, $h + g_0$, or F on S is required.

1.2 Exact initialized algorithm

Use $g = g_0 + I_D$ and the original ambient, undamped B-adaPG, equations (2.3) and (2.6) of Ou–Latafat–Themelis, arXiv:2508.01353v2. Start at arbitrary $x_0 \in C \cap D$ and arbitrary $\gamma_0, \gamma_1 > 0$. The first update uses γ_1, and every update is

$$x_{k+1} = \operatorname*{argmin}_{y \in S} \left\{ g_0(y) + \langle \nabla f(x_k), y - x_k \rangle + \frac{D_h(y, x_k)}{\gamma_{k+1}} \right\}.$$

For $k \geq 1$ and distinct consecutive iterates define

$$\rho_k = \gamma_k/\gamma_{k-1}, \quad r_{k+1} = \sqrt{1 + \rho_k}, \quad \delta_{k+1} = 2r_{k+1}, \quad H_k = \nabla h - \gamma_k \nabla f,$$

$$\ell_k = \frac{\Delta_f(x_k, x_{k-1})}{\Delta_h(x_k, x_{k-1})}, \qquad \alpha_k = \frac{D_h(x_k, x_{k-1})}{D_h(x_{k-1}, x_k)},$$

$$\Lambda_k = \frac{2D_{h^*}\big(\nabla h(x_k) + \delta_{k+1}[H_k(x_k) - H_k(x_{k-1})], \nabla h(x_k)\big)}{\delta_{k+1}^2 \Delta_h(x_k, x_{k-1})},$$

$$\rho_{k+1} = \min\left\{ r_{k+1}, \frac{\alpha_k}{(1 + \alpha_k) 2r_{k+1}[\Lambda_k - (1 - \gamma_k \ell_k)]_+} \right\}, \qquad \gamma_{k+1} = \rho_{k+1}\gamma_k.$$

Here $\Delta_v(a, b) = D_v(a, b) + D_v(b, a)$, $[t]_+ = \max(t, 0)$ and positive divided by zero is $+\infty$. The conjugate h^* and all controller vectors are full ambient quantities; neither projection nor kernel restriction is allowed. If consecutive iterates coincide, stop before evaluating zero denominators and extend the sequence constantly.

2 Convergence theorem

Theorem 1. *Under the hypotheses of Section 1, each proximal update is uniquely defined in $C \cap D$, and every nonstopped generated stepsize is finite and positive. For arbitrary $x_0 \in C \cap D$ and $\gamma_0, \gamma_1 > 0$, the exact orbit satisfies*

$$F(x_k) \to F(z), \qquad x_k \to z, \qquad D_h(z, x_k) \to 0, \qquad \nabla f(x_k) \to \nabla f(z).$$

No stepsize floor or ceiling, damping, boundary avoidance, bounded constraint normals, affine-f hypothesis, continuity of h on K, or strict boundary-kernel geometry is added. No rate is asserted. General closed convex constraints, including lower-dimensional ones, are allowed.

3 Well-posedness and stopping

For source applicability, extend f outside U by

$$\widehat{f}(x) = \sup_{u \in U} \{f(u) + \langle \nabla f(u), x - u \rangle\}.$$

This is proper closed convex and equals f on U, so it changes neither a feasible value nor an algorithm gradient. A proximal update is equivalent to minimizing

$$h(y) + \gamma g_0(y) + I_D(y) + \langle \gamma \nabla f(x) - \nabla h(x), y \rangle.$$

The objective is proper, lower semicontinuous and supercoercive, since g_0 has an affine minorant and h is 1-coercive. Thus it attains a minimum. Since the open set C meets D, it meets $\operatorname{ri} D$. The exact convex sum rule applies, and essential smoothness gives $\operatorname{dom} \partial h = C$. A minimizer therefore lies in $C \cap D$; strict convexity of h there makes it unique. With $g = g_0 + I_D$, select

$$\xi_j = \frac{\nabla h(x_{j-1}) - \nabla h(x_j)}{\gamma_j} - \nabla f(x_{j-1}) \in \partial g(x_j). \tag{1}$$

The conjugate h^* is finite and continuously differentiable on all of $\mathbb{R}^n$. The positive definite continuous primal Hessian and inverse-function theorem also make h^* twice continuously differentiable with positive definite Hessian. All ambient controller evaluations are valid. At distinct consecutive iterates both kernel divergences are positive. The first selector branch is positive and finite; the second is positive or infinite. Hence every generated stepsize is finite and positive.

If two consecutive iterates coincide, (1) proves optimality. Conversely, an interior feasible minimizer is locally D-optimal and satisfies $0 \in \nabla f(x) + \partial g(x)$, so its next iterate coincides. Thus either the stopped sequence is constantly z from some point on, or every finite iterate has $F(x_j) > F(z)$. We consider the latter case below.

4 Exact energy with the smooth defect retained

For $w \in S$ write $P_j(w) = F(x_j) - F(w)$. For $k \geq 2$ set

$$d_k = 1 - \frac{\rho_k}{2r_k}, \qquad U_k(w) = D_h(w, x_k) + \gamma_k(1 + r_k)P_{k-1}(w) + d_k D_h(x_k, x_{k-1}).$$

Energy arguments start at $k = 2$: no selector inequality is imposed on the arbitrary initial ratio $\rho_1 = \gamma_1/\gamma_0$. For the actual selected subgradient in (1), define

$$\widetilde{D}_g(u, x_j) = g(u) - g(x_j) - \langle \xi_j, u - x_j \rangle,$$

$$\widetilde{D}_F(u, x_k) = F(u) - F(x_k) - \langle \nabla f(x_k) + \xi_k, u - x_k \rangle.$$

These quantities are nonnegative at feasible arguments, including with the constraint indicator. Put $\vartheta = r_{k+1}$ and

$$T_k = (1 - \gamma_k \ell_k)\Delta_h(x_k, x_{k-1}), \qquad E_{k+1} = \rho_{k+1}\langle H_k(x_k) - H_k(x_{k-1}), x_{k+1} - x_k \rangle.$$

The optimality relation and three-point identity give

$$\begin{aligned} D_h(w, x_{k+1}) + \gamma_{k+1}P_k(w) + D_h(x_{k+1}, x_k) &= D_h(w, x_k) + E_{k+1} - \gamma_{k+1}Q_k(w), \\ P_{k-1}(w) - P_k(w) &= T_k/\gamma_k + \widetilde{D}_F(x_{k-1}, x_k), \end{aligned}$$

where

$$Q_k(w) = D_f(w, x_k) + \widetilde{D}_g(x_{k+1}, x_k) + \widetilde{D}_g(w, x_{k+1}).$$

For the second identity use $\nabla f(x_k) + \xi_k = -[H_k(x_k) - H_k(x_{k-1})]/\gamma_k$. Adding $\gamma_{k+1}\vartheta$ times that identity to the first yields

$$\begin{aligned} &D_h(w, x_{k+1}) + \gamma_{k+1}(1 + \vartheta)P_k(w) + D_h(x_{k+1}, x_k) \\ &= D_h(w, x_k) + \gamma_{k+1}\vartheta P_{k-1}(w) - \rho_{k+1}\vartheta T_k + E_{k+1} \\ &\quad - \gamma_{k+1}\{Q_k(w) + \vartheta \widetilde{D}_F(x_{k-1}, x_k)\}. \end{aligned}$$

This is the merit identity underlying Lemma 3.1 of [1], with its nonnegative terms displayed. Fenchel's inequality in Bregman form,

$$\langle v, y - x \rangle \le D_h(y, x) + D_{h^*}(\nabla h(x) + v, \nabla h(x)),$$

applied to $v = \delta_{k+1}[H_k(x_k) - H_k(x_{k-1})]$ gives

$$E_{k+1} \le \frac{\rho_{k+1}}{\delta_{k+1}} D_h(x_{k+1}, x_k) + \frac{\delta_{k+1}\rho_{k+1}}{2}\Lambda_k \Delta_h(x_k, x_{k-1}).$$

This uses the full ambient conjugate and the exact positive dual displacement of the stated controller. Substitution of $\delta_{k+1} = 2r_{k+1}$, collection of coefficients, and retention of the smooth defect give

$$\begin{aligned} U_{k+1}(w) \le U_k(w) &- A_k P_{k-1}(w) - B_k D_h(x_k, x_{k-1}) \\ &- \gamma_{k+1} D_f(w, x_k), \end{aligned} \tag{2}$$

where

$$A_k = \gamma_k(1 + r_k) - \gamma_{k+1} r_{k+1},$$

$$B_k = d_k - \rho_{k+1} r_{k+1} \frac{1+\alpha_k}{\alpha_k} [\Lambda_k - (1 - \gamma_k \ell_k)].$$

The bracket in B_k is signed. The first branch gives $\rho_k \le r_k$ and $d_k \ge 1/2$ for $k \ge 2$, and

$$A_k \ge \gamma_k (r_k - \rho_k) \ge 0.$$

If the bracket is nonpositive, $B_k \ge d_k$. If positive, the second branch bounds the subtracted term by $1/2$. In either case $B_k \ge 0$. At $w = z$, all $P_j(z) \ge 0$, so $U_k(z)$ is nonnegative and nonincreasing. Therefore

$$D_h(z, x_k) \le U_2(z), \qquad \sum_{k \ge 2} \gamma_{k+1} D_f(z, x_k) < \infty. \tag{3}$$

5 Boundedness and an optimal cluster point

We first make the valid interior-anchor reverse-coercivity argument explicit. For $a \in C$, choose $r > 0$ with $\overline{B}(a, r) \subset C$, and put $M = \max_{\overline{B}(a,r)} h$. Evaluating the supporting inequality at $a + rp/\|p\|$, where $p = \nabla h(x) \ne 0$, gives

$$D_h(a, x) \ge r\|p\| + h(a) - M. \tag{4}$$

A bounded reverse divergence to an interior anchor bounds p; continuity of ∇h^* on all of $\mathbb{R}^n$ puts $x = \nabla h^*(p)$ in a compact subset of C. If $z \in C$, apply this to (3). If $z \in K \setminus C$, the assumed reverse-sublevel boundedness applies instead. Thus the orbit is bounded. No boundary point-separation property of D_h is used.

We next prove

$$\inf_{j \ge 0} F(x_j) = F(z). \tag{5}$$

Suppose to the contrary that $b = \inf_j F(x_j) > F(z)$. For sufficiently small $t > 0$, the point $w = (1-t)z + tx_0$ is in $C \cap D$ and satisfies $F(w) < b$. Put $d = b - F(w) > 0$. Then $P_j(w) \ge d$, so (2) bounds $D_h(w, x_j)$ and confines the orbit to a compact subset of C by (4). Telescoping from $m = 2$ and retaining the terminal objective term gives

$$d \left[\sum_{k=m}^{N} A_k + \gamma_{N+1}(1 + r_{N+1}) \right] \le U_m(w).$$

The bracket is exactly

$$\sum_{k=m}^{N+1} \gamma_k + \gamma_m r_m.$$

Consequently $\sum \gamma_k < \infty$ and $\gamma_k \to 0$.

Here is the needed contradiction directly from the exact selector. A slightly enlarged compact convex neighborhood $E \subset C$ of the orbit has Hessian bounds $0 < \mu I \preceq \nabla^2 h \preceq$

$M_h I$. Proximal comparison with the old iterate, boundedness of ∇f, and local Lipschitz continuity of finite convex g_0 give

$$\frac{\mu}{2}\|x_k - x_{k-1}\|^2 \le D_h(x_k, x_{k-1}) \le B\gamma_k\|x_k - x_{k-1}\|.$$

Thus consecutive differences tend to zero. Relative smoothness and the upper Hessian bound make ∇f Lipschitz on a neighborhood of this compact set: $(LM_h/2)\|x\|^2 - f$ is convex there, and the convex smoothness equivalence applies. Let $t_k = x_k - x_{k-1}$, $M_k = \nabla^2 h(x_k)$ and $p_k = \nabla h(x_k)$. Uniform Taylor expansions imply

$$p_k - p_{k-1} = M_k t_k + o(\|t_k\|), \qquad \Delta_h(x_k, x_{k-1}) = t_k^T M_k t_k + o(\|t_k\|^2), \qquad \alpha_k \to 1.$$

With $R_* = \max\{\rho_1, (1+\sqrt{5})/2\}$, the first branch implies $\rho_k \le R_*$ and $1 \le r_{k+1} \le R_*$ for all relevant indices. Thus δ_{k+1} is bounded above and below. Also $0 \le \ell_k \le L$, and Lipschitz continuity of ∇f gives

$$v_k := H_k(x_k) - H_k(x_{k-1}) = M_k t_k + o(\|t_k\|).$$

On a compact dual neighborhood, $\nabla^2 h^*(p_k) = M_k^{-1}$; uniform expansion gives

$$2D_{h^*}(p_k + \delta_{k+1} v_k, p_k) = \delta_{k+1}^2 v_k^T M_k^{-1} v_k + o(\|t_k\|^2).$$

Therefore $\Lambda_k \to 1$, while $\gamma_k \ell_k \to 0$. The positive part in the second branch tends to zero, so that branch tends to $+\infty$. Eventually the first branch is selected: $\gamma_{k+1} = \sqrt{1+\rho_k}\gamma_k > \gamma_k$. This contradicts $\gamma_k \to 0$ and proves (5).

No finite iterate in the nonstopped run is optimal. Hence (5) supplies indices tending to infinity with objective values tending to $F(z)$. Boundedness, closedness of S, continuity of $F = f + g_0$ on S, and uniqueness imply that this subsequence tends to z. We have obtained an optimal cluster point, not yet optimality of every cluster.

6 Feasible-anchor estimate and scalar separation

Let Q be the intersection of S with a sufficiently large closed ball containing z and the orbit. It is compact and convex. For an anchor $a \in Q$ and an update $x = x_k$, $y = x_{k+1}$, (1) and the three-point identity give

$$D_h(a, y) - D_h(a, x) \le -D_h(y, x) + \gamma_{k+1}\{\langle \nabla f(x), a - y\rangle + g_0(a) - g_0(y)\}. \tag{6}$$

Indeed, the exact sum rule splits the selected subgradient into one from g_0 and a normal $n \in N_D(y)$; its contribution $\langle n, a - y\rangle$ is nonpositive. No norm bound on normals is needed. Boundedness of ∇f and g_0 on Q supplies a uniform constant $T > 0$ bounding the braces above. Holding $a = x_p$ fixed and telescoping yields, for $q > p$,

$$D_h(x_p, x_q) \le T \sum_{i=p}^{q-1} \gamma_{i+1}. \tag{7}$$

Choose a ball strictly containing Q and its relative-smoothness constant $L > 0$. Since both orbit endpoints are interior,

$$D_f(x_p, x_q) \le LD_h(x_p, x_q) \le LT \sum_{i=p}^{q-1} \gamma_{i+1}. \tag{8}$$

This is precisely where uniformity up to bounded boundary approaches is used.

Define the continuous nonnegative scalar function $a(x) = D_f(z, x)$ on Q. If $u, v \in Q$ and $D_f(u, v) = 0$, the affine support of f at v touches at u. It is consequently a subgradient of f at u; differentiability on the open set U gives $\nabla f(u) = \nabla f(v)$. Moreover $f(u) - f(v) = \langle \nabla f(v), u - v \rangle$, so $a(u) = a(v)$. Compactness and continuity therefore imply, for every $\eta > 0$ with a nonempty indicated set,

$$m(\eta) := \min\{D_f(u, v) : u, v \in Q,\ |a(u) - a(v)| \ge \eta\} > 0. \tag{9}$$

Thus a scalar change of size η from index p to q costs cumulative stepsize at least $m(\eta)/(LT)$. This does not assert separation of primal points by boundary Bregman distance.

7 Cumulative-step excursions and common cluster gradients

We prove $a(x_k) \to 0$. If $\sum \gamma_k < \infty$, equations (8)–(9) show that $a(x_k)$ is Cauchy: any fixed nonzero scalar difference would require a positive cumulative tail. Its subsequence tending to z has scalar limit zero, proving the assertion.

Now suppose $\sum \gamma_k = \infty$. If the assertion fails, some $\eta > 0$ has arbitrarily late indices p with $a(x_p) \ge 2\eta$. If $a(x_i) \ge \eta$ for every $i \ge p$, the infinite stepsize tail contradicts (3). Otherwise let $q > p$ be the first index with $a(x_q) < \eta$. For every old endpoint $p \le i < q$ one has $a(x_i) \ge \eta$, whereas $|a(x_p) - a(x_q)| \ge \eta$. Consequently

$$\sum_{i=p}^{q-1} \gamma_{i+1} a(x_i) \ge \frac{\eta m(\eta)}{LT} > 0$$

by (8)–(9). Arbitrarily late such intervals contradict the vanishing tails of the convergent series in (3). Thus in both cases $D_f(z, x_k) \to 0$.

Every cluster point y therefore satisfies $D_f(z, y) = 0$. The same touching-support argument, now at z, gives

$$\nabla f(y) = c := \nabla f(z), \qquad f(y) - f(z) = \langle c, y - z \rangle. \tag{10}$$

Compactness and continuity imply

$$e_k := \nabla f(x_k) - c \to 0. \tag{11}$$

No strictness assumption was used in obtaining a common gradient for all clusters.

8 Unique-minimizer endpoint trap

Set $G(x) = g_0(x) + \langle c, x\rangle$. The point z minimizes G on S. Indeed, f and g_0 are continuous in a neighborhood of z, so the exact sum rule applied to $f + g_0 + I_S$ gives $c + \xi + n = 0$ for some $\xi \in \partial g_0(z)$ and $n \in N_S(z)$. Their supporting inequalities imply $G(x) \geq G(z)$ for $x \in S$. This argument uses N_S directly and does not split the normal of $K \cap D$. For every cluster point y, (10) gives

$$G(y) - G(z) = F(y) - F(z). \tag{12}$$

Thus a G-optimal cluster must equal z by uniqueness of the F-minimizer.

Put $W_k = D_h(z, x_k)$. Equation (6) with $a = z$ becomes

$$\begin{aligned} W_{k+1} - W_k \leq & -D_h(x_{k+1}, x_k) - \gamma_{k+1}[G(x_{k+1}) - G(z)] \\ & + \gamma_{k+1}\langle e_k, z - x_{k+1}\rangle. \end{aligned} \tag{13}$$

For every $\varepsilon > 0$, there are $\delta > 0$ and N such that

$$k \geq N, \quad W_{k+1} \geq \varepsilon \quad \Longrightarrow \quad G(x_{k+1}) - G(z) \geq \delta. \tag{14}$$

Otherwise one could choose a convergent subsequence of late endpoints with $W \geq \varepsilon$ and vanishing G-gap. By (12) its limit is z, and the assumed boundary compatibility $D_h(z, y_i) \to 0$ when $y_i \to z$ contradicts $W \geq \varepsilon$. If high-W endpoints eventually do not occur, the implication is vacuous and the desired bound already holds.

Let R_0 bound $\|z - x_k\|$ and increase N until $R_0\|e_k\| \leq \delta/2$. At every late endpoint with $W_{k+1} \geq \varepsilon$, (13) implies $W_{k+1} < W_k$. Along the optimal subsequence $x_p \to z$, boundary compatibility gives $W_p \to 0$. Choose such a late p with $W_p < \varepsilon$. A first crossing back to $W_q \geq \varepsilon$ would force $W_q < W_{q-1} < \varepsilon$, a contradiction. Hence

$$D_h(z, x_k) \to 0. \tag{15}$$

We have not inferred point convergence from this reverse-distance limit.

9 Large-step endpoints and objective convergence

Dropping the nonpositive displacement term from (13) and using $W_{k+1} \geq 0$ gives

$$0 \leq G(x_{k+1}) - G(z) \leq \frac{W_k}{\gamma_{k+1}} + R_0\|e_k\|. \tag{16}$$

For each fixed $\tau > 0$, equations (11) and (15) show that G-gaps tend to zero along all endpoint indices j with $\gamma_j \geq \tau$. Every cluster along these indices equals z, by (12) and uniqueness. Compactness thus gives $x_j \to z$ and $F(x_j) \to F(z)$ along all such indices.

Proximal comparison with x_{j-1} and relative descent on the bounded orbit give

$$F(x_j) \le F(x_{j-1}) - (1/\gamma_j - L)D_h(x_j, x_{j-1}). \tag{17}$$

Every step with $\gamma_j \le 1/L$ is objective-nonincreasing. If there are infinitely many indices with $\gamma_j > 1/L$, their endpoint values tend to $F(z)$ by (16). Between successive such endpoints all intermediate values are at most the value at the preceding large-step endpoint. The lower bound $F \ge F(z)$ then proves convergence of the full objective sequence.

If only finitely many indices have $\gamma_j > 1/L$, the objective is eventually nonincreasing. Its limit must be $F(z)$: a larger limit, together with strict suboptimality of every finite-prefix value in a nonstopped run, would contradict (5). In both cases $F(x_k) \to F(z)$. Every cluster of the bounded orbit lies in closed S and is optimal by continuity. Uniqueness forces every cluster to equal z, so $x_k \to z$. This proves the theorem through objective convergence, without boundary Bregman point separation.

10 A nonlinear example without any strictness on the closure

Put $e = 1/256$, $a = 1/100$, $K = [-1,1] \times [0,e]$, and $C = \operatorname{int} K$. With $0 \log 0 = 0$, define

$$\begin{aligned} b(x) &= (1+x)\log(1+x) + (1-x)\log(1-x), \\ q(y) &= -\sqrt{y} - \sqrt{e-y}, \\ h(x,y) &= y[b(x) - ax\sin(\log y)] + q(y) \quad (y>0), \\ h(x,0) &= -\sqrt{e}, \end{aligned}$$

and set $h = +\infty$ off K. For arbitrary $\beta, \kappa > 0$, take $f(x,y) = y + \beta y^2$, $g_0(x,y) = \kappa(x+1)$ and $D = \mathbb{R}^2$. The kernel is constant on the entire bottom edge, while the forward objective is nonlinear. Continuity of b on $[-1,1]$ and $y \sin\log y \to 0$ make h continuous on K. Its infinity extension is proper lower semicontinuous, and its bounded domain gives 1-coercivity. Write $t = \log y$, $B = b''(x) = 2/(1-x^2)$ and $b'(x) = \log((1+x)/(1-x))$. On C,

$$h_{xx} = yB, \quad h_{xy} = b'(x) - a(\sin t + \cos t),$$

$$h_{yy} = q''(y) - ax(\cos t - \sin t)/y, \quad q''(y) = \tfrac{1}{4}[y^{-3/2} + (e-y)^{-3/2}].$$

Putting $u = \operatorname{atanh} x$, the inequality $\cosh u \ge \sqrt{2}|u|$ gives $b'(x)^2/B \le 1$. Since $B \ge 2$,

$$h_{xy}^2/B \le 2 + 2a^2, \qquad y h_{yy} \ge 4 - 2a.$$

Let $c_0 = 2 - 2a - 2a^2 > 0$. The Schur complement satisfies

$$h_{yy} - h_{xy}^2/h_{xx} \geq c_0/y \geq c_0/e > 0. \tag{18}$$

Thus the Hessian is positive definite. Continuity extends convexity to boundary chords. The derivatives

$$h_x = y[b'(x) - a\sin t], \quad h_y = b(x) - ax(\sin t + \cos t) + q'(y)$$

have divergent norm at every approach to the boundary: horizontal-edge divergence comes from q', with bounded remaining terms, and vertical-edge divergence away from horizontal edges comes from b'. Inward one-sided difference quotients are $-\infty$ on every edge, also at corners. Hence there are no boundary subgradients, and strict interior convexity establishes Legendreness.

For fixed $(X, Y) \in K$ and $(u, v) \to (X, Y)$ from C, continuity reduces Assumption 2.4(i) to

$$h_x(u, v)(X - u) \to 0, \qquad h_y(u, v)(Y - v) \to 0.$$

The first follows from $(X - u)b'(u) \to 0$ at vertical endpoints and bounded v. The second follows from bounded nonsingular terms and the square-root limits $(Y - v)q'(v) \to 0$ at $Y = 0, e$. These cover mixed corners. Reverse sublevels are bounded because K is bounded, proving Assumption 2.4(ii).

For $f = y + \beta y^2$, the only nonzero Hessian entry is $f_{yy} = 2\beta$. Equation (18) implies $L\nabla^2 h - \nabla^2 f \succeq 0$ throughout C whenever $L \geq 2\beta e/c_0$. Thus the stronger uniform relative smoothness holds globally, not merely along an assumed orbit. The finite function $g_0 = \kappa(x + 1)$ is affine. On K, $F = y + \beta y^2 + \kappa(x + 1) \geq 0$, with equality only at $(-1, 0)$. All hypotheses therefore hold. On the bottom edge, h, g_0, $h + g_0$, and F are affine; none is strictly convex on K. In particular, even strict convexity of g_0 on the h-flat segments fails. Nevertheless, every exactly initialized orbit converges to $(-1, 0)$. Since every finite iterate has positive y, no finite iterate is optimal or invokes stopping. No explicit orbit formula, controller-branch inheritance, or rate is asserted.

11 Scope of the result

Under the same boundary-uniform relative-smoothness hypothesis, this theorem removes the strict-g_0 assumption used by a split-curvature argument. It also includes, but does not need, strict convexity of $h + g_0$ on S, or strict convexity of g_0 on each nontrivial h-affine segment, provided the unique-minimizer and other stated hypotheses hold.

The source's compact-interior relative-smoothness condition allows the modulus to deteriorate as the boundary is approached. Our condition fixes one modulus throughout every bounded interior region, including such approaches. It is used essentially both in

the cumulative-step estimate (8) and in the common small-step threshold in (17). The unrestricted unique-minimizer boundary problem under the source's weaker smoothness assumption remains unresolved by this argument. No general boundary separation property of h is asserted, and no claim of novelty or priority is made.

References

[1] H. Ou, P. Latafat and A. Themelis. Linesearch-free adaptive Bregman proximal gradient for convex minimization under local relative smoothness. *arXiv:2508.01353v2*, 5 June 2026. `https://arxiv.org/abs/2508.01353v2`.